

A New Classification Approach of Cultural Heritage: Deep Learning Based Image Recognition

Serhat Atas

Computer Engineering Department

University of Firat (FU)

Elazig, Turkey

serhatats99@gmail.com

Mehmet Karaköse

Computer Engineering Department

University of Firat (FU)

Elazig, Turkey

mkarakose@gmail.com

Abstract— With the development of technology, artificial intelligence is becoming gigantic and entering many areas. This development of artificial intelligence has also revealed beautiful studies by making contributions to the cultural heritage. These studies have shown that our cultural self and technology are actually not far from each other and are instrumental in useful studies. Many platforms have contributed to the development of various studies as a front for combining cultural heritage with developing artificial intelligence. In these various studies, not only artificial intelligence, but many different technologies have been used and tried to contribute to the cultural heritage. We have brought together deep learning and cultural heritage by deconstructing a study in this field. In our study, together with the special data set we have created, we have carried out a study that tries to recognize cultural heritage structures with deep learning by choosing a suitable model for our study, and we have described the contributions we have made in this study in this article.

Keywords— cultural heritage, deep learning, image classification

I. INTRODUCTION

Although the great development in the field of deep learning has resulted in the latest technological applications, as we have seen, it has enabled us to come together with our cultural heritage items left over from the past, to preserve our past, to revive it, and to carry out various Deciphering studies on these objects, resulting in successful studies. We can't just Decipher deep learning as an example of these studies, because over time our cultural selves have started to come together with different software languages and technologies to survive. As an example, historical museums have been animated via a browser resource and online museum browsing has been enabled. In another study, the museum sightseeing event was made into 3-dimensional so that people could really feel within those historical objects. In addition, as we mentioned at the beginning in this study, many studies have emerged in the field of deep learning, such as detecting the wear of cultural heritage objects, completing the damaged areas with artificial intelligence, estimating how old cultural objects are. In this article, we will explain the method used in our study and reveal the experimental results we

have obtained. In the method we have used, we aim to detect cultural heritage objects at a high rate from the images of cultural heritage objects that we will provide as test data from the model we have trained to recognize cultural heritage objects. In the model training and experimental results, we use a special data set consisting of images of 10 different cultural heritage objects.

Casillo et al. carried out a cloud-based study with IoT devices to protect cultural heritage structures and found a useful study by using GAN architecture. In this way, they managed to identify future errors and wear and tear in cultural heritage structures [1]. Franca et al. have developed a microservice study to facilitate access to cultural heritage and have revealed a study that facilitates access to information about cultural heritage objects to users on different platforms, including mobile and web, by providing communication through these microservices. They used different software languages (Java, Angular, Ionic) in their studies [2]. Shimoni et al. by detecting the damages caused by the ground and caused by the year in the cultural heritage structures, they carried out a study that determined the damage status of the structures from many factors on the objects [3]. Kambau et al. conducted a study using CNN and RNN to classify cultural heritage items [4]. Rashid et al. processed big data by using big data tools and cloud base and worked on creating archives for the protection of cultural heritage [5]. Madani et al. created humanoid robotics by signing a different study and designed this robot to answer people's cultural heritage questions [6]. Belhi et al. carried out a study aimed at classifying, completing and explaining damaged data in order to protect cultural heritage data [7]. Hess et al. applied different approaches by trying to visualize cultural heritage structures digitally [8]. Alam et al. carried out a study on the cultural heritage of Bangladesh that uses CNN and RNN together to add subtitles to Decals [9]. Using the XRF method, Chopp et al. have carried out a large-scale study on cultural heritage objects to eliminate noise and remove unnecessary signals [10]. Kulkarni et al. with deep learning methods, they conducted a study that implemented the recognition of images given via browser in order to digitize the cultural heritage of India

[11]. Monna et al. using deep learning, they conducted a study that identified cultural heritage items built in Indonesia from satellite images [12]. Sperli has made a deep learning based chatbot for cultural heritage [13]. Cui et al. they carried out a study using deep learning models in order to determine the damage to cultural heritage items [14]. Llamas et al. by performing a similar study to the one we have done, they compared performance using different models with deep learning techniques in order to classify historical buildings [15]. Matrone et al performed a segmentation process on 3D cultural heritage images [16]. Hatir et al. performed the decomposition process using deep learning on historical stone works [17]. Pierdicca et al have carried out a segmentation process on cultural heritage items. The architecture of their study is given in Figure 1 below. [18].

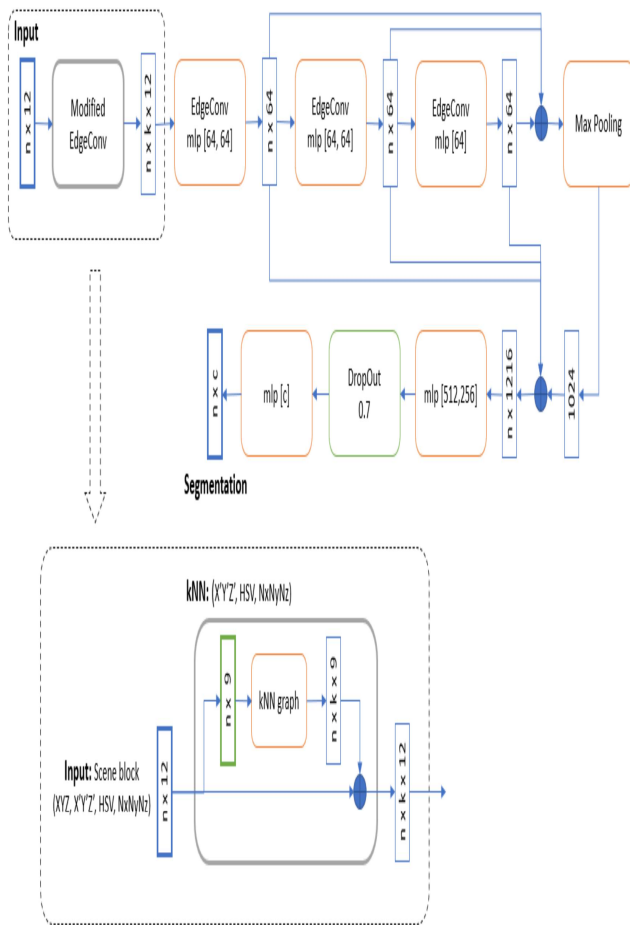


Figure 1. Pierdicca et al. block diagram of their work

II. PROPOSED APPROACH

The proposed approach was made using CNN (Convolutional Neural Network). In our study, we allocate 80% of the data set belonging to 10 different cultural heritage objects that we have obtained specifically as education and 20% as test data. With the training data we have allocated, we train the EfficientNetB0 model in a 5-step way. After training our model, we expect that by giving the test data we have allocated to our trained model, it will provide recognition from 10 different cultural heritage objects. The block diagram of our system is given in Figure 2 below.

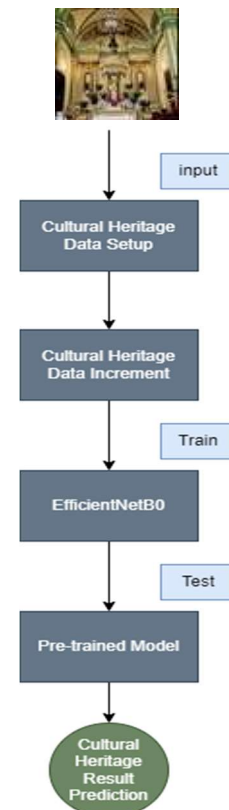


Figure 2. Block diagram of our system for recognizing Cultural Heritage objects

As can be understood from Figure 2, the visual data we received as input are divided into 80%-20%, train and test, in the Cultural Heritage Setup Data section, then in the Cultural Heritage Data Increment process, the train data we have is entered into training with the EfficientNetB0 model, we enter a sample test data into the trained model and reach the final stage.

A. Cultural Heritage Setup Data

Our special data set that we obtained in our study consists of 12000 images. At this first stage, we are separating the data set so that we can use it as a train and test in other steps. In this separation process, we separated 80% of our data as train data according to the needs of the model, and used the rest as test data. This means that the train data in our data consists of 9600 images and the test data consists of 2400 images.

B. Cultural Heritage Data Increment

After the data in our study has passed through the first stage, we enable the data to be reproduced and the model to recognize from different angles by performing operations such as rotation, expansion, upgrade, zoom on the visuals in order for the model to be fed better. This stage allows us to increase the accuracy rate of the model on images given in different sizes and varieties.

C. Model Training

At this stage, we are now taking the train data we obtained last and giving it to the EfficientNetB0 model as input, then we train the model. In another study Alirezazadeh et al. worked with the EfficientNetB0 model, made comparisons with different models and proved the accuracy of the EfficientNetB0 model [19].

The EfficientNetB0 model architecture is given below in Figure 3 [20].

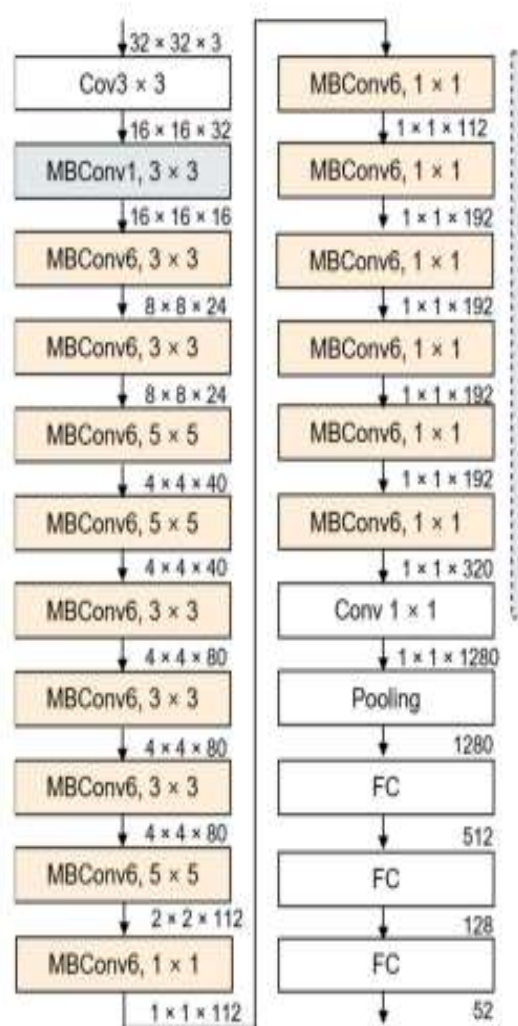


Figure 3. EfficientNetB0 Architecture [20]

The structure of our model is given in Figure 4 below. This output is a summary that shows the layers of a model and how many parameters there are in each layer. The InputLayer is the place where data will be given to the model as input, and it has no parameters. The Data Augmentation layer is a Sequential model that contains a series of data augmentation operations used to augment educational data. This layer has no parameters. The EfficientNetB0 layer is a Functional model that contains the weights of the EfficientNetB0 model. There are 4,049,571 parameters in this layer. The GlobalAveragePooling2D layer reduces its dimensions by taking global averages of the outputs of the EfficientNetB0 model. This layer has no parameters. The output layer provides outputs for 10 classes in a classification problem with 10 classes. There are 12,810 parameters in this layer. In total, there are 4,062,381 parameters in the model, and only 12,810 of them are trainable (parameters at the output_layer layer). The remaining parameters are the pre-trained weights of the EfficientNetB0 model and are not updated during training.

Model: "model"		
Layer (type)	Output Shape	Param #
input_layer (InputLayer)	[(None, 224, 224, 3)]	0
data_augmentation (Sequential)	(None, None, None, 3)	0
efficientnetb0 (Functional)	(None, None, None, 1280)	4049571
global_avg_pool_layer (GlobalAveragePooling2D)	(None, 1280)	0
output_layer (Dense)	(None, 10)	12810
Total params: 4,062,381		
Trainable params: 12,810		
Non-trainable params: 4,049,571		

Figure 4. Model Structure

D. Test

By inserting a sample test data into the trained model we have obtained, we expect it to recognize which cultural heritage class the object belongs to from the image we have given.

III. EXPERIMENTAL RESULTS

We obtained the experimental results we obtained by working on the Google Colaboratory. Our data set in our study has been created by collecting images from 10 different types of cultural heritage. This data set contains 12000 images, and 9600 of these images were used for model training, and 2400 of them were used as test data to obtain our experimental results. The dimensions of the images in our data set are between 2KB-10KB and each of our images has been revised Decently as 64*64. Below, our train data set is given in Table I in detail with its classes. In the same way, the test data are given in Table II.

TABLE I.
TRAIN DATA SET

Train Data Set	Number of Image
Altar	880
Apse	950
Bell Tower	960
Column	860
Dome(inner)	1104
Dome(outer)	1006
Flying Buttress	978
Vault	942
Gargoyle	900
Stained Glass	1020
Total Images	9600

TABLE II.
TEST DATA SET

Test Data Set	Number of Image
Altar	224
Apse	256
Bell Tower	188
Column	301
Dome(inner)	231
Dome(outer)	215
Flying Buttress	193
Vault	312
Gargoyle	200
Stained Glass	280
Total Images	2400

We train our model with the train data set that we have given in the form of table above, but since we are doing data replication operations in addition to this data set, our model trains on more images than the data set we have provided. The graph of the loss and accuracy ratios that we received as a result of the model training is given in Figure 5 below.

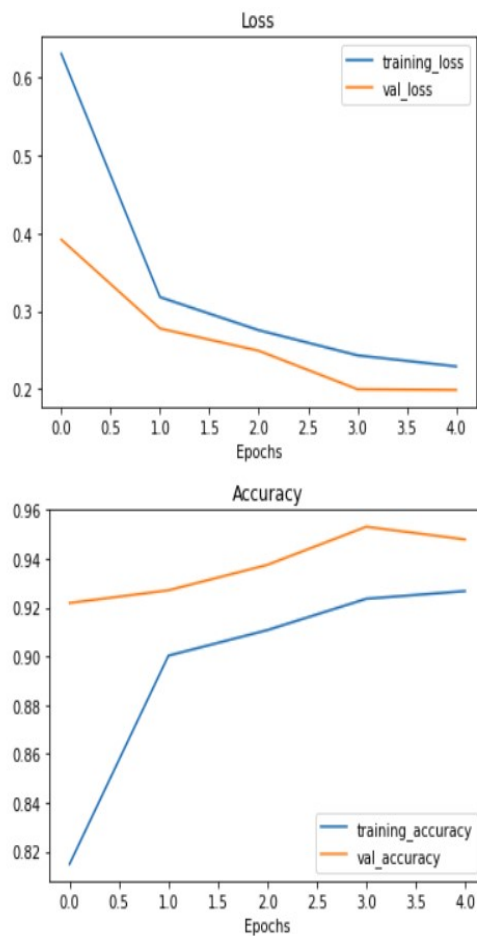


Figure 5. Loss and accuracy ratios for training and validation

On the model, with the test data mentioned above, we take the accuracy rates of the classes separately by making a detection operation on the images belonging to each cultural heritage class and calculate the accuracy rate of our model. The model accuracy rate is given in detail in Table III below.

TABLE III.
MODEL PERFORMANCE

Test Data Set	Model Performance in Test Images
Altar	%95.07
Apse	%76.92
Bell Tower	%92.39
Column	%93.77
Dome(inner)	%94.89
Dome(outer)	%95.56
Flying Buttress	%87.02
Vault	%96.02
Gargoyle	%94.42
Stained Glass	%96.24
Total Performance	%92.23

The experimental results we obtained in our study are also given in Figure 6 below in graphic form.

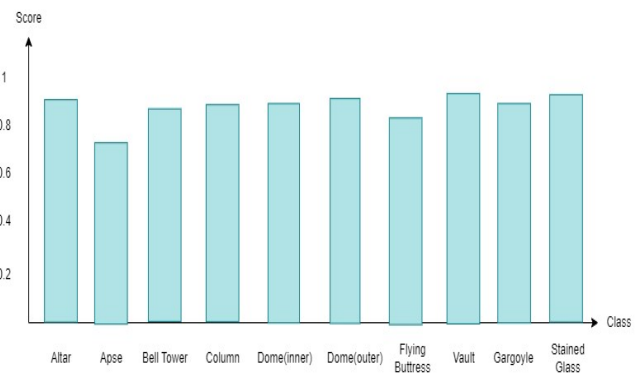


Figure 6. Model detection accuracy ratio graph

Sample images of the estimates we obtained as a result of the study are given in Figures 7,8 and 9 below. By uploading the images in our test file, we run the code and get the following prediction results.



Figure 7. Sample output



Figure 8. Sample output



Figure 9. Sample output

In addition to all these, the comparison of the study we have done with the studies in the literature is given in Table IV. When we look at the table, we observe that our study contributes to the field of cultural heritage in a different way from other studies, with the model and special data set we have used.

TABLE IV.
COMPARISON OF STUDIES

Reference	Technique	Result
Franca ve ark. [2]	FullStack	Web Site
Kambau ve ark. [4]	CNN-RNN	Classification
Rashid ve ark. [5]	Big Data Tools	Archive
Alam ve ark. [9]	CNN-RNN	Image Subtitle
Choop ve ark. [10]	XRF	Noise and Signal Removal
Our Study	CNN	Cultural Heritage Image Classification

IV. CONCLUSION

As a result of the work we have done, we have created a recommended method on deep learning by trying to emphasize the value of cultural heritage. We have carried out a study in this field together with our special data set that we have created by combining deep learning and cultural heritage. Dec.

In our study, we used a CNN-based deep learning algorithm to identify cultural heritage objects according to the class in which they are located. As it can be understood from the study we have done, we have achieved better results with the special data set we have created and the data increase we have made before introducing the model into the training. We have proven the accuracy of the EfficientNetB0 model that we used in our study and that it has reached high values in the detection rate.

In our study, we tried to contribute to the field of cultural heritage by obtaining a 92.23% accuracy rate on the experimental results we made in 10 different cultural heritage classes.

In addition to all these, we tried to remind that the cultural heritage area is of great importance for us and our land, it can be intertwined with technology.

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